YESWANTH KUMAR KALUVA

Software Engineer — Backend & Full-Stack Development

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PROFESSIONAL SUMMARY

Final-year Computer Science (Data Science) student (CGPA 8.55/10.0) with three internships spanning backend, full-stack, and general software engineering work. Experienced across the complete software development lifecycle — design, coding, testing, debugging, and Agile team collaboration — building production-style systems including a 33-table relational schema, REST APIs, authentication/authorization logic, and automated workflow systems across multiple platforms and technology stacks. Solved 500+ data structures and algorithms problems; strong analytical and problem-solving foundation with practical exposure to system design. Adaptable to remote, hybrid, and in-person team environments.

TECHNICAL SKILLS

- Programming Languages: Python, Java, JavaScript, C, SQL
- Core Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, Software Testing & Debugging, REST Architecture, System Design fundamentals
- Backend & APIs: Node.js, Express.js, Flask, REST API Design, Authentication & Authorization (JWT), Role-Based Access Control (RBAC), Microservices concepts
- Databases: MySQL, SQLite, Schema Design, Query Optimization, Indexing, Database Transactions (ACID), Connection Pooling
- Frontend: React.js, Vite, HTML5, CSS3, JavaScript (ES6+), Responsive Web Design
- Platforms & Tools: Docker, Azure Blob Storage, Google Cloud Platform (Cloud Run), Git, GitHub, Linux, VS Code, Postman, CI/CD fundamentals
- Practices: Agile, Scrum, Structured Logging & Monitoring, Unit & Integration Testing, Cross-functional collaboration

EXPERIENCE

Backend Engineering Intern (Part-time, On-site) March 2026 – May 2026

nDMatrix, Hyderabad

- Designed and coded a 33-table relational MySQL schema for a multi-module digital evaluation platform, covering submission, evaluation, and annotation workflows across three user roles.
- Built and debugged REST APIs in Node.js and Express, implementing role-based access control via reusable middleware to enforce authorization checks before business logic executed.
- Implemented JWT-based authentication with token-replay protection, testing edge cases across authenticated request flows.
- Engineered multi-step database transactions (BEGIN/COMMIT/ROLLBACK) to guarantee atomic writes and eliminate data integrity issues on partial failures.
- Added structured logging and monitoring hooks to speed up debugging and issue diagnosis in a production-style environment.
- Collaborated with frontend and backend engineers across Agile sprints on feature planning, code integration, and cross-team debugging.

Software Engineering Intern (Part-time, On-campus) August 2025 – July 2026

Vignan University, Guntur

- Designed and developed secure multi-stage workflow backends for leave management, payroll processing, and complaint handling, reducing manual processing errors by approximately 60%.
- Built a GitHub-style attendance heatmap dashboard for internal academic systems, improving administrative tracking visibility by approximately 40%.
- Implemented REST-based API integrations and workflow progress indicators, improving approval turnaround speed by approximately 35%.
- Added structured logging and debugging checkpoints to identify and resolve issues during backend testing and workflow execution.
- Partnered with faculty stakeholders, frontend developers, and backend engineers across Agile development cycles to test and ship iterative feature releases.

Full-Stack Development Intern June 2024 – July 2024

Fusil Solutions, Hyderabad

- Designed and coded a full-stack recipe-sharing web application using Flask, MySQL, HTML, CSS, and JavaScript, including authentication and secure content workflows.
- Implemented RESTful APIs supporting full CRUD operations, a content moderation pipeline, and an admin analytics dashboard.
- Applied structured input validation and backend checks, testing form submissions to improve data integrity.
- Tested backend APIs using Postman, debugging and logging issues during integration testing to support release quality.

PROJECTS

Real-Time Speech Emotion Recognition *(Python, TensorFlow, CNN, LSTM, Flask, REST API)*

- Designed, coded, and tested a CNN-LSTM deep learning model, achieving approximately 85% classification accuracy on multi-class speech datasets.
- Built and debugged a Flask-based inference service handling concurrent audio inputs with sub-300ms average latency.
- Implemented preprocessing pipelines and REST API endpoints; performed iterative testing and hyperparameter tuning to balance accuracy with inference speed.

Hospital Management System *(Flask, MySQL, REST APIs)*

- Designed and coded RESTful backend services for appointment scheduling and doctor workflow management.
- Optimized SQL queries and schema structure, reducing average data retrieval latency by approximately 40%.
- Implemented and tested role-based access controls and notification workflows.

EDUCATION

B.Tech in Computer Science and Engineering (Data Science) Expected May 2026

Vignan's Foundation for Science, Technology and Research, Guntur | CGPA: 8.55 / 10.0

- Intermediate (MPC), Narayana Junior College, Vijayawada — 97.5% | 2022
- Secondary School Certificate, Oxford High School, Podili — 100% | 2020

ACHIEVEMENTS AND CERTIFICATIONS

- Solved 500+ algorithmic problems across LeetCode, HackerRank, and GeeksforGeeks covering arrays, trees, graphs, and dynamic programming.
- Ranked Top 5 in Department Hackathon for Full-Stack Application Development.
- Consistently ranked among the top academic performers in the Data Science department.
- Google Cybersecurity Certificate — Coursera (2022).
- NPTEL Certifications: Data Analytics Using Python, Pattern Recognition, Responsible AI and Safe Systems.